DESIGN OF 6-LANE FLEXIBLE PAVEMENT BY USING IIT PAVE SOFTWARE

Project Guide :
T. NARESH
(ASST. PROFESSOR of
Civil Engineering)

Project by: **Batch D12**
19RJ5A0161 - T HARSHAVARDHAN

ABSTRACT

- Road transport plays a vital role in social, economic and industrial development of the country. Roads are used by different vehicle class which results in failure of the pavement due to early deterioration. The failure of bitumen pavement is due to fatigue cracking and rutting deformation.

- This paved a way for developing the software like IITPAVE which determines the stress and strains values at critical location at different layer of pavement. This project aims at collection of data related to traffic studies and subgrade soil CBR values from the field and designing the flexible pavement as per the guidelines of IRC: 37-2018. Later it is analyzed using the IITPAVE Software to check whether it satisfies the specific requirements.

- This project briefly describes the use of IITPAVE software for project analysis of National Highway NH365BB (Suryapeta – Khammam) with chainage length of 10km from Mothey, Suryapet to Nayakulagudem, Kothagudem Telangana.

1. INTRODUCTION

1.1 General

The design of flexible pavement depends on different factors that affect the strength and CBR value of the subgrade soil such as load, properties of soil, environmental and climatic factors. Using flexible pavement is necessary to ensure road safety in a large city and will help to save time and avoid difficulties and errors which might happen during the design. Pavements are constructed as per IRC guidelines. The main reason for failure of bitumen pavement is due to excessive strain and deformation at critical locations in the pavement.

1.2 Definition of Road

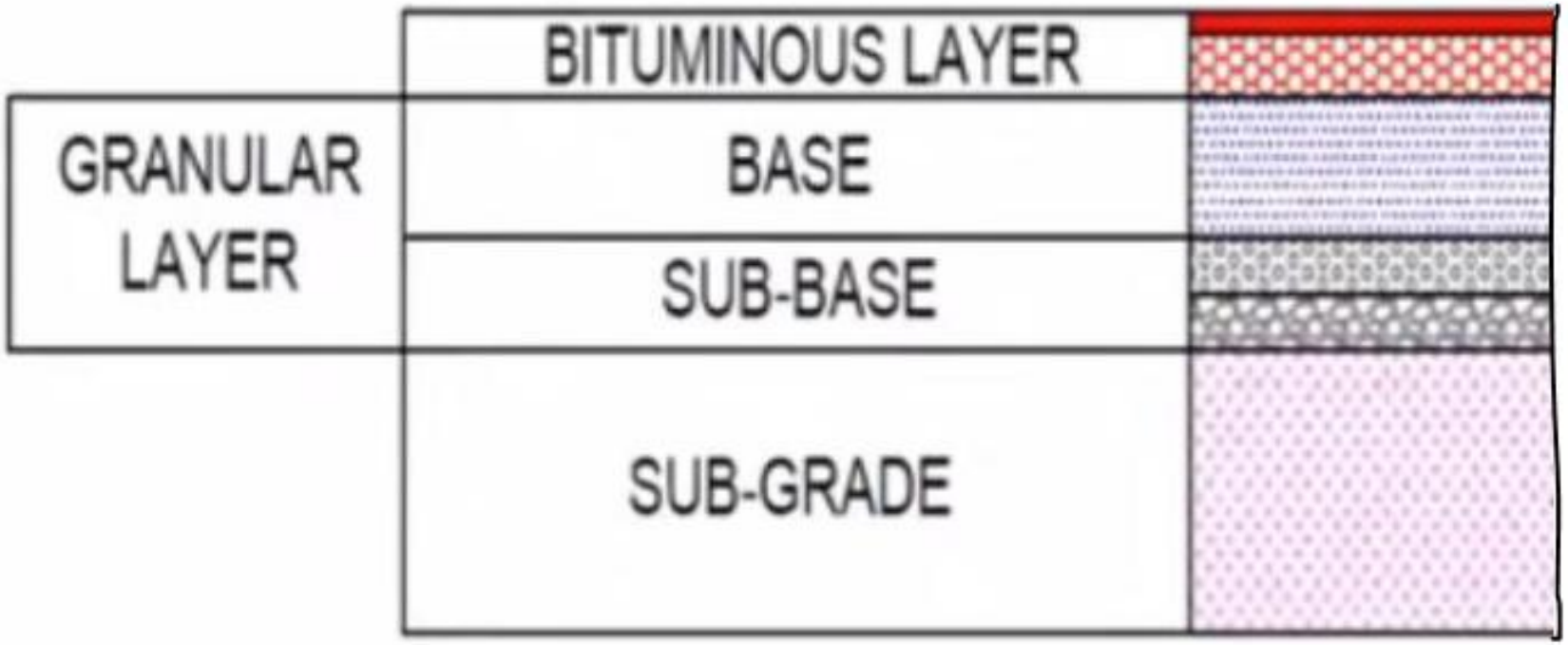
A wide way leading from one place to another, especially one with a specially prepared surface which vehicles can use.

1.3 Composition of Road Structure

Road Structure Cross Section is composed of the following components

1. Sub Base
2. Base Course
3. Sub Grade
4. Surface/Wearing Course

1. INTRODUCTION



1. INTRODUCTION

FUNCTION OF LAYERS:

I Surface (or wearing) course:
This is the topmost layer: its function is to provide a smooth, strong, abrasion.
Resistant and reasonably impervious course. Since it is directly in contact with the vehicle tyres, it has to resist the imposed wheel loads and transmit them safely to the layer below.

II. Base course:
This is immediately below the surface course and its function is to distribute the stresses transmitted throught the surface course evenly onto the layers below.
Invariably . it consists of granular or bituminous material, and acts as a structural part of the pavement.

III. Sub-base course:
This comes just below the base course and provides additional help to the courses above it in distributing the load. It also help in preventing soil grains of the sub grade from intruding into the base course above, and counteracts frost action ,I f any it consist of stabilized soil or soil aggregate mixes, which facilitate drainage of free water from the pavement.

iv. Sugrade:
it is act as a foundation for the highway.

1. INTRODUCTION

1.4 Types of Pavements

***1.4.1 Pavement Definition*:** Pavement is the actual travel surface especially made durable and serviceable to withstand the traffic load commuting upon it. Pavement grants friction for the vehicles thus providing comfort to the driver and transfers the traffic load from the upper surface to the natural soil. All hard road pavements usually fall into two broad categories namely

a. Flexible Pavement,

b. Rigid Pavement.

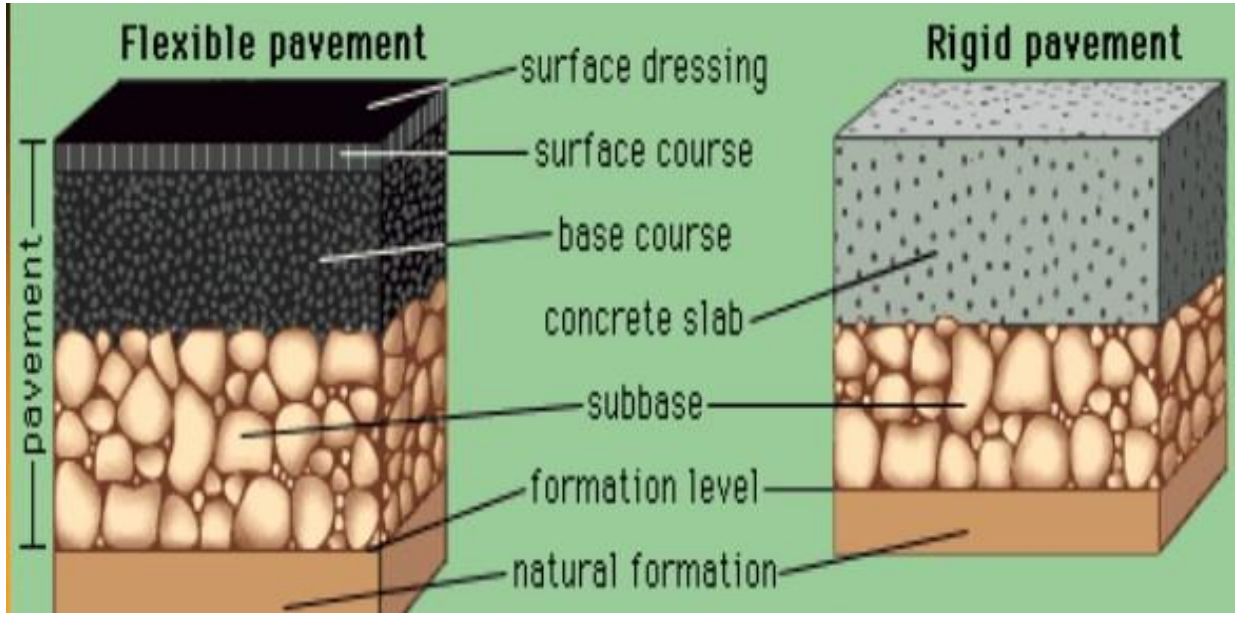


Figure 1.1 typical-detail-of-road-section

1. INTRODUCTION

1.4.5 Requirements of a pavement

- Sufficient thickness to distribute the wheel load stresses to a safe value on the sub grade soil.
- Structurally to strong to withstand all types of stress imposed upon it.
- Adequate coefficient of friction to prevent skidding of vehicles.
- Smooth surface to provide comfort to road users even at high speed.
- Produce least noise from moving vehicles.
- Dust proof surface so that traffic safety is not impaired by reducing visibility

.

- Impervious surface, so that sub-grade soil is well protected and long design life with low maintenance cost.

1. INTRODUCTION

1.6 IIT PAVE For Pavement Analysis

- IIT PAVE software is an improved version of FPAVE which is developed by research scheme R-56 of MORTH.
- This is a multilayer analysis programmer used for design, analysis of the flexible or bitumen pavement using IRC: 37-2018 guidelines.
- In this software we enter thickness of pavement layers, loads applied over the surface of pavement, tire pressure, spacing between the wheels and Poisson´s ratio as inputs.
- After running the software actual horizontal tensile strain and vertical compressive strains at critical locations of the pavement are obtained as output.

1. INTRODUCTION

1.7 Study Area

- Design of 6 lane pavement for the project analysis of National Highway NH365BB (Suryapeta – Khammam) with chainage length of 10km from Mothey, Suryapet to Nayakulagudem, Kothagudem Telangana.

- National Highway 365BB is a national highway in India. It is a secondary route of National Highway 65. NH-365BB traverses the states of Telangana and Andhra Pradesh in India. It starts at Suryapet and ends at Kovvur (Rajamahendravaram).

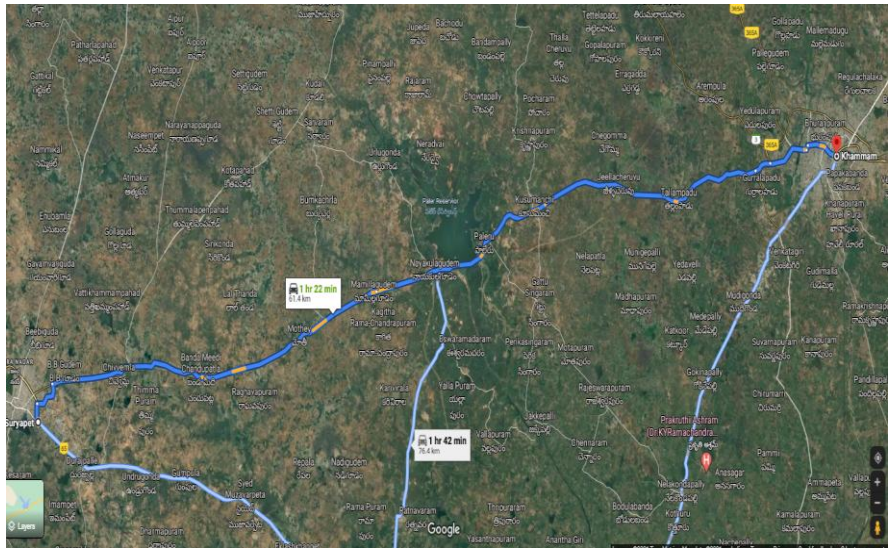


Figure 1.3 Satellite Map for the NH-365BB from Suryapet to Khammam

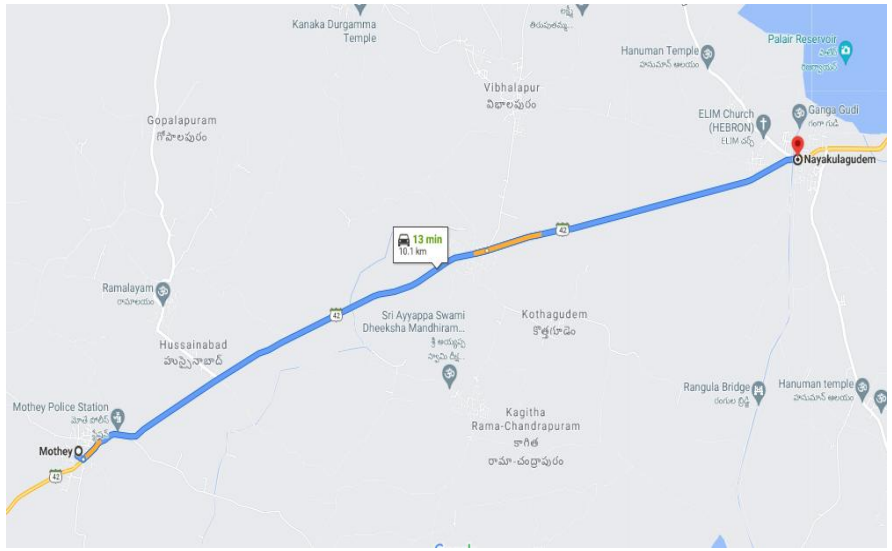


Figure 1.4 Map for Mothey to Nayakulagudem

2. LITERATURE REVIEW

Harish G R (4) (2017), the design of flexible pavement by using cementitious materials for base and sub-base layers for crack relief and cementitious base and sub-base with SAMI (Stress Absorbing Membrane Interface) as per the guidelines of IRC: 37-2012 codebook and later the same pavement is analyzed in the IIT PAVE software. IIT PAVE is the multi-layer analysis programmer used for analysis of flexible pavement and to determine the stresses and strains at critical locations of the pavement. In this case the road stretch is selected around Bangalore and engineering properties of subgrade soil has been studied. From their study it is concluded that the pavement which is designed by using cementitious base and sub-base with SAMI will give better serviceability because of substantial reduction in strains and thickness, and also it is cost effective compared to other pavement materials.

Komershetty Gouthami, Dr KSR Murthy, D. Sandya Rani (7) (2017), the life cycle cost of different pavement compositions. For this purpose, they have collected the information associated to vehicular traffic studies, axle load survey and sub-grade CBR values of a selected road. The project road considered for study is Solapur to Sangareddy section of NH-9. From the obtained data, the design of flexible pavement is carried out using IRC: 37-2012 guidelines. Later a roadway is analyzed in IIT PAVE software and compared with the pavement which is designed by the alternative materials available in the IRC: 37-2012 with the conventional granular materials. Based on the results obtained in design of pavement using IRC: 37-2012 and analysis carried out using IIT PAVE software, they have concluded that the alternate pavements with less total pavement thickness can cater the same traffic as that of conventional pavement except the alternate pavement with CTB and GSB.

3. OBJECTIVE OF THE STUDY

- Traffic Analysis by using collected traffic volume count data by adopting suitable CVPD values.
- Pavement design based on average daily traffic as per the IRC: 37 -2018 guidelines and analyzing pavement in IITPAVE software.
- To design and evaluate the performance of fatigue and rutting performance of flexible pavements using IITPAVE Software.
- To design of various pavement Layer thickness with the help of IITPAVE Software.
- To determine the best engineering and economic alternatives for the road sections.

4. METHODOLOGY

- The flexible pavement is designed by an empirical method, as per the IRC guidelines (IRC: 37-2018) which is based mainly on CBR value of subgrade soil and design charts to determine the total pavement thickness. The design of flexible pavement is done according to IRC: 37-2018 guidelines and it is analyzed in IIT PAVE software. The road selected for this project is Suryapet – Khammam section of NH-365BB.

- A flexible pavement is modelled as an elastic multilayer structure. The mechanistic empirical software called IITPAVE is used to analyze the pavement. The stress analysis IITPAVE software is used to compute the stresses and strains at critical location of the pavement. Horizontal Tensile strain at the bottom of the bitumen layer and Vertical Compressive strain on the top of the subgrade are considered as the critical parameters to limit cracking and rutting in bituminous and non-bituminous layers.

- The allowable tensile and compressive strains at critical location of the pavement are calculated using empirical equations mentioned in IRC: 37-2018 guidelines which is later compared with the actual tensile and compressive strains as computed by IIT PAVE software.

5. EXPERIMENTAL WORK

5.1 General

The experimental work carried out by 2 phases those are mentioned in below;

1. Laboratory and field Investigation of soil
2. CVPD based traffic volume survey

5.2 Field Investigation of soil

Detailed field investigation comprising soil condition survey, in-situ testing and sampling was carried out on the selected homogeneous section.

5.2.1 In-situ Moisture content

Figure 5.1: Rapid Moisture Meter

Table 5.1: Field Moisture test result

Field Moisture content (%)	**8%**

5. EXPERIMENTAL WORK

5.2.2 In-situ Density by Core Cutter method

Figure 5.2: Core Cutter

Table 5.2: Core Cutter test results

Water Content (%)	**12%**
Dry Density (g/cc)	1.75

5. EXPERIMENTAL WORK

5.3 Laboratory Investigations of soil

5.3.1 Laboratory Tests on Soil

To evaluate the properties of the existing subgrade the laboratory tests conducted are as follows:

1. Grain Size Analysis
2. Atterberg Limits
3. Free Swell Index
4. Compaction and Optimum Moisture Content
5. California Bearing Ratio Test

5. EXPERIMENTAL WORK

5.3.1.1 Grain Size Analysis

Figure 5.3: Grain Size Analysis

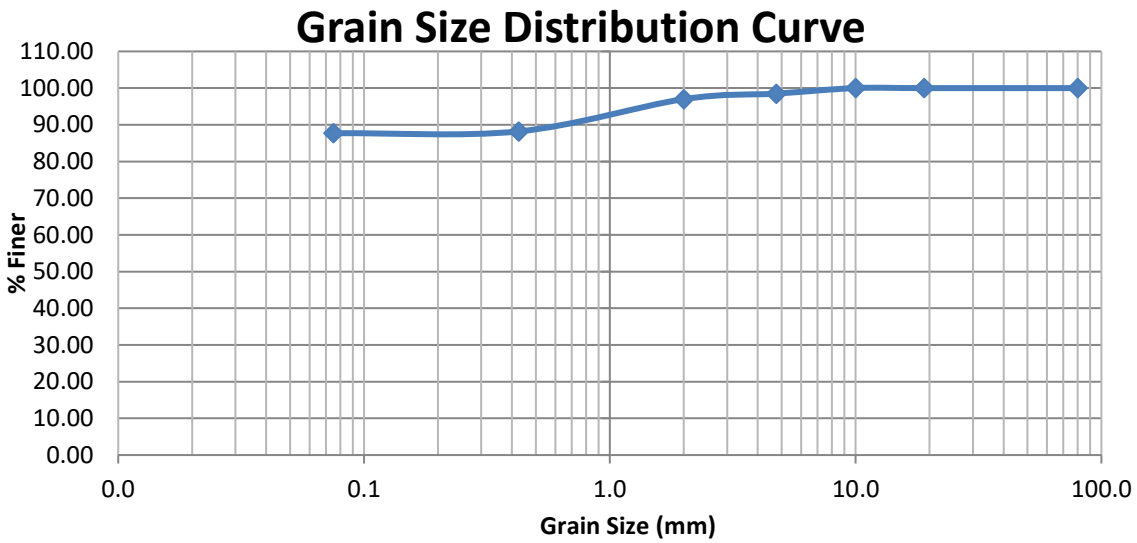


Figure 5.4: Grain Size Distribution Curve

5. EXPERIMENTAL WORK

5.3.1.2 Atterberg Limits

Figure 5.5: Liquid limit test

Test result: 31.8%

Figure 5.6: Plastic Limit

Test result: 16.9%

Plasticity Index = LL – PL = 31.8% - 16.9% = 14.9%

5. EXPERIMENTAL WORK

5.3.1.4 Compaction Test

Figure 5.7: Soil Compactor

Figure 5.8: Soil compaction

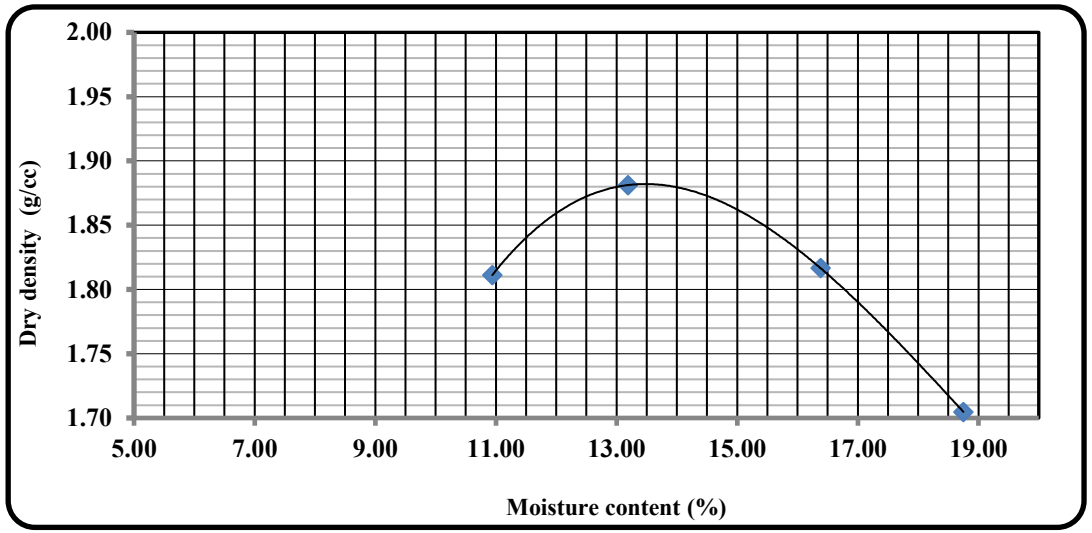


Figure 5.9: Dry Density Vs Moisture Content

5. EXPERIMENTAL WORK

5.3.1.5 California Bearing Ratio Test

Figure 5.10: CBR Moulds

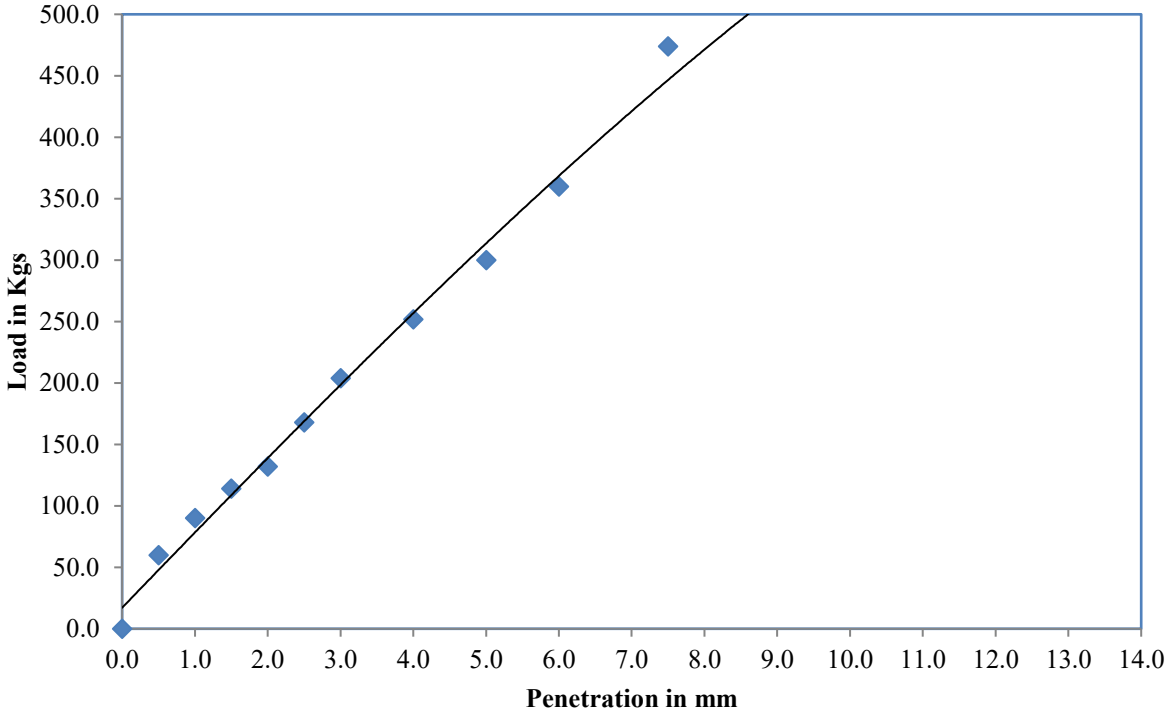


Figure 5.11: Sample Calculation Graph of CBR

5. EXPERIMENTAL WORK

Existed soil CBR values : Collect existing / Embankment CBR Data as per SP 19 one sample for every one kilometer length. Compute 90th percentile CBR value for high volume roads and 80th percentile CBR value for low volume roads. For this study we considered 10km road. The test results of soaked CBR values given below;

Table 5.8 CBR Test results

Location (Chainage) (km)	4 days soaked CBR (%)
0	4.2
1	5.5
2	5.1
3	4.6
4	6.7
5	6.5
6	5.3
7	6.1
8	7.1
9	6.6

Table 5.9 Compute 90th percentile existing CBR value

Chainage (km)	4 days soaked CBR	Count	Decending order CBR	Percentile
0	4.2	10	4.2	100
1	5.5	9	4.6	90
2	5.1	8	5.1	80
3	4.6	7	5.3	70
4	6.7	6	5.5	60
5	6.5	5	6.1	50
6	5.3	4	6.5	40
7	6.1	3	6.6	30
8	7.1	2	6.7	20
9	6.6	1	7.1	10

5. EXPERIMENTAL WORK

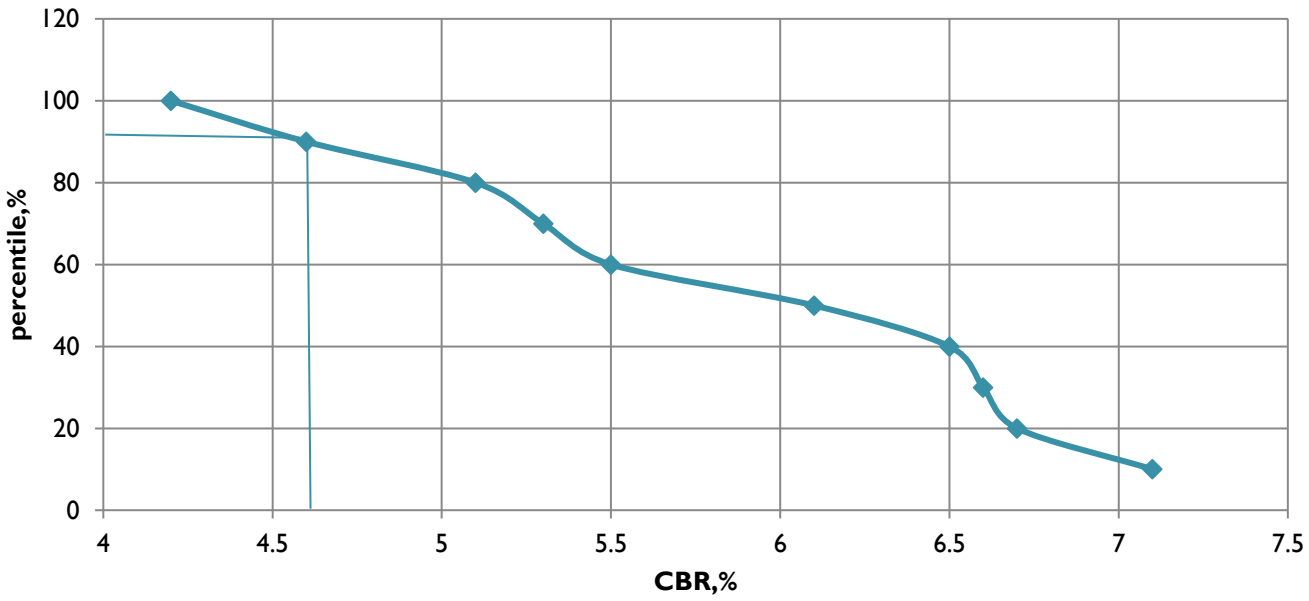


Figure 5.12: 90th percentile CBR value

Result: 90th percentile CBR value found as **4.6% &** CBR value of borrow soil is 8.5 %

5. EXPERIMENTAL WORK

5.4 Traffic volume

5.4.1 Commercial Vehicle Per Day: Based on traffic volume count survey carried out on project corridor as per IRC SP 19-2001 for 7days 24 hour. Average of 7days traffic consider as average daily traffic (ADT). Only commericial vehicles (having the laden weight of 3 Tonnes or more) count to be considered.

Table 5.10: Commercial Vehicle Per Day

DAY	CVPD (Two ways)
Monday	2502
Tuesday	2365
Wednesday	2743
Thursday	2100
Friday	2268
Saturday	2598
Sunday	2643
Average	2459.85 ≈ 2460

Result: Initial CVPD based on traffic volume survey = 2460 (Two ways)

6. IITPAVE SOFTWARE

- IITPAVE software is used to calculate the Actual Horizontal Tensile Strain in Bituminous layer and Actual Vertical Compressive Strain on sub-grade.

- The actual strains are computed using various trial pavement thickness combinations. The tyre pressure used in the analysis is 0.56 MPa. Standard axle used is dual type and single tyre load of 20,000 N. The Poisson's ratio of bituminous layer is taken as 0.35.

- The E values and Poisson's ratio obtained from above are given as inputs to the software and strains are calculated.

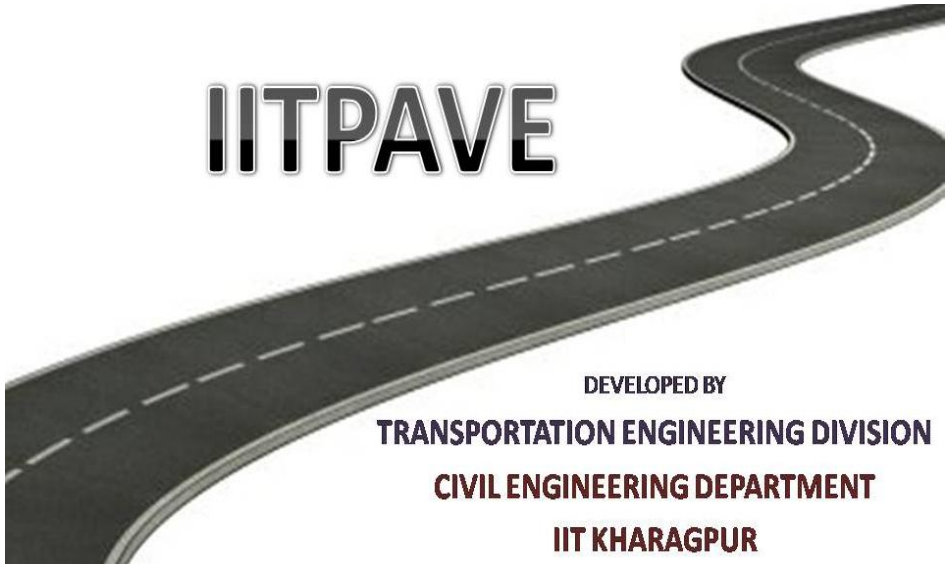


Figure 1.2: IIT PAVE

6. IITPAVE SOFTWARE

6.2 INPUTS TO THE SOFTWARE:

- First, number of layers to be selected from drop down menu to fix up input boxes for different layer.
- Next, Elastic modulus (E) values of the various layers in MPa, Poisson's ratio and thickness of the layers in mm excluding the subgrade thickness are to be provided.
- Single wheel load and the tyre pressure are to be provided (tyre pressure of 0.56 MPa has been used for calibration of the fatigue equation and the same pressure can be used for stress analysis. Change of pressure even up to 0.80 MPa has a small effect upon stress values in lower layers.)
- Then the number of points for stress computations is to be given through the drop down menu for Analysis points.
- Then corresponding to different points, the values of depth Z in mm and the corresponding value of radial distances from wheel centre (r) in mm are to be provided.
- Provide whether analysis is for single wheel load or double wheel load by clicking 1 or 2 and 2 will be the most common case.

7. DESIGN AND ANALYSIS OF PAVEMENT

7.3 Design of Pavement (IRC 37: 2018 and IIT PAVE)

The design steps for pavement given below;

1. Compute subgrade effective resilient modulus (M_{RS})
2. Compute design traffic in terms of MSA (N_{DES})
3. Select layer combination
4. Compute permissiable horizontal & vertical strain
5. Compute actual strain as per IITPAVE.(try combinations of different thicknesses)
6. Selection of Economical design

7. DESIGN AND ANALYSIS OF PAVEMENT

7.3.1.1 **Compute subgrade effective resilient modulus (M_{RS}):**

M_{RS} (MPa) = 10 * CBR for CBR <= 5

= 17.6 * $(CBR)^{0.64}$ for CBR > 5

M_{RS} = modulus of subgrade soil.

From the **5.3.1.5** in chapter 5 : 90th percentile CBR value found as **4.6% &** CBR value of borrow soil is **8.5 %.**

Existing soil M_{RS} (Mpa) = 10 * (CBR) = 10 * 4.6 = 46 Mpa

Borrow soil M_{RS} (Mpa) = 17.6 * $(CBR)^{0.64}$ = 17.6 * $(8.5)^{0.64}$ = 69.237 Mpa

Wheel load 40000Newtons & tyre pressure 0.56

Then submit these inputs in IITPAVE and the strains are obtained as shown in Figure 7.3(a & b).

7. DESIGN AND ANALYSIS OF PAVEMENT

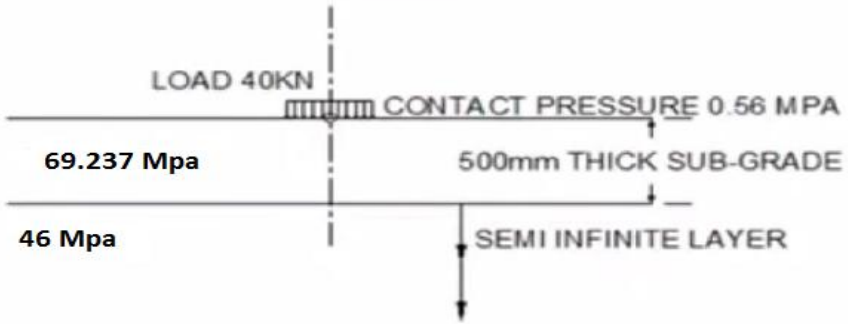


Figure 7.1 resilient modulus values with respect to the layer

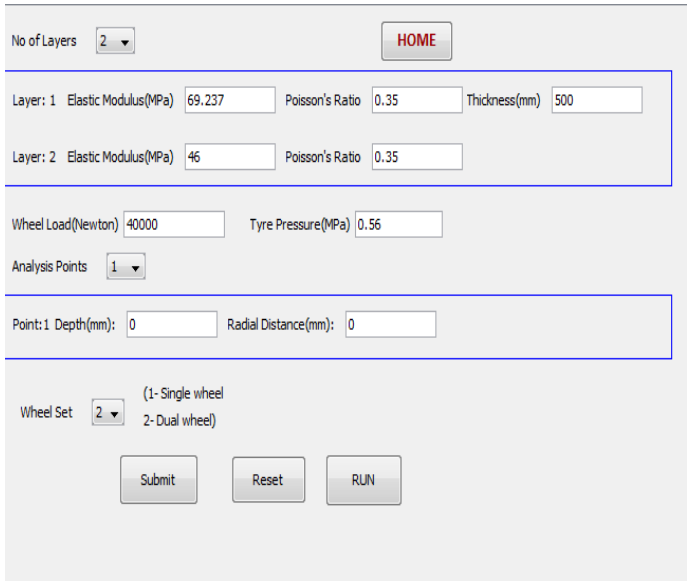


Figure 7.3(a) Input file

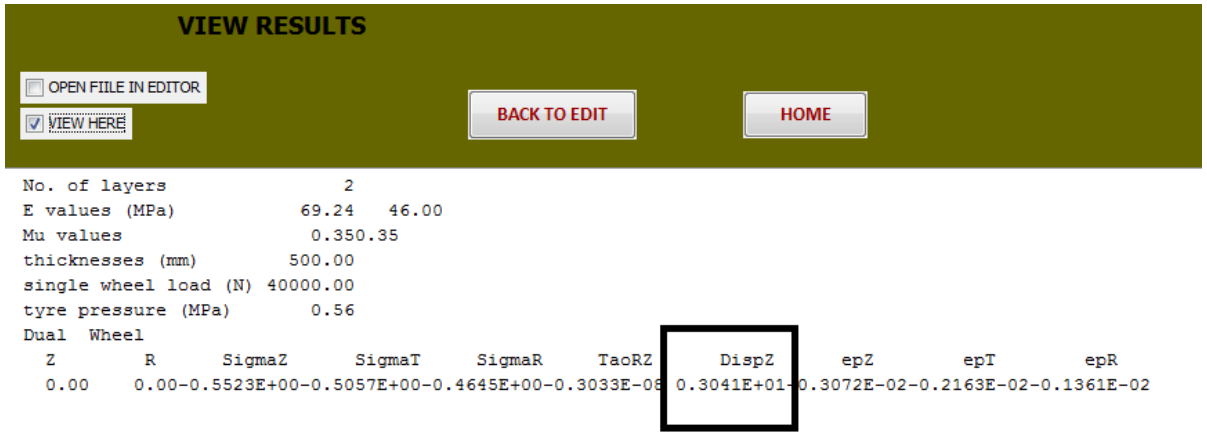


Figure 7.3(b)Computed deflection

7. DESIGN AND ANALYSIS OF PAVEMENT

$M_{RS} = [2(1 - \mu^2)\ pa\ ] / \delta$

Where,
p = contact pressure = 0.56 MPa
a = radius of circular contact area, which can be calculated using the load applied (40,000 N) and the contact pressure 'p' (0.56 MPa) = 150.8 mm
μ = Poisson's ratio
$M_{RS} = [2(1 - \mu^2)\ pa\ ] / \delta = [2(1-0.35^2\ ) \times 0.56 \times 150.8] / 3.041 = 48.736$ Mpa

Figure 7.4 resilient modulus for combined borrow soil & existed soil

7. DESIGN AND ANALYSIS OF PAVEMENT

7.3.1.2 Design Traffic

$$N_{Des} = [(365\ ((1+r)^n - 1) / r\ )) * A * D * F]$$

Data for design:

Dual Three lane carriageway

Initial CVPD based on traffic volume survey = 2460 CVPD (two ways) = P

Traffic growth rate = 5%

Vehicle damage factor = 5 (same for both direction)

Construction period = 1 year

Design period = 20years

Derive Initial traffic in year of construction:

$A = P(1+r)^x = 2460\ (1+0.05)^1 = 2583$ CVPD (two ways) ≈ 2584 CVPD = 1292(assume 50 % in each side)

Derive MSA:

$N_{DES} = [(365\ ((1+r)^n - 1) / r\ ))\ x\ A\ X\ D\ X\ F]$

$= [(365\ (1+0.05)^{20} - 1) / 0.05\ )\ x\ 1292\ X\ 0.6\ X\ 5]$ = 46.7797 MSA = Say **47MSA**

7. DESIGN AND ANALYSIS OF PAVEMENT

7.3.1.3 Layer Selection

Existence soil soaked CBR value of 4.6% & 47 MSA

Figure 12.1 Catalogue for pavement with bituminous surface course with granular base and sub-base - Effective CBR 5% (Plate-1)

GSB = 200mm
WMM = 250mm
BASE / BINDER COURSE = 140mm
SURFACE COURSE = 40mm
SUBGRADE = 500mm

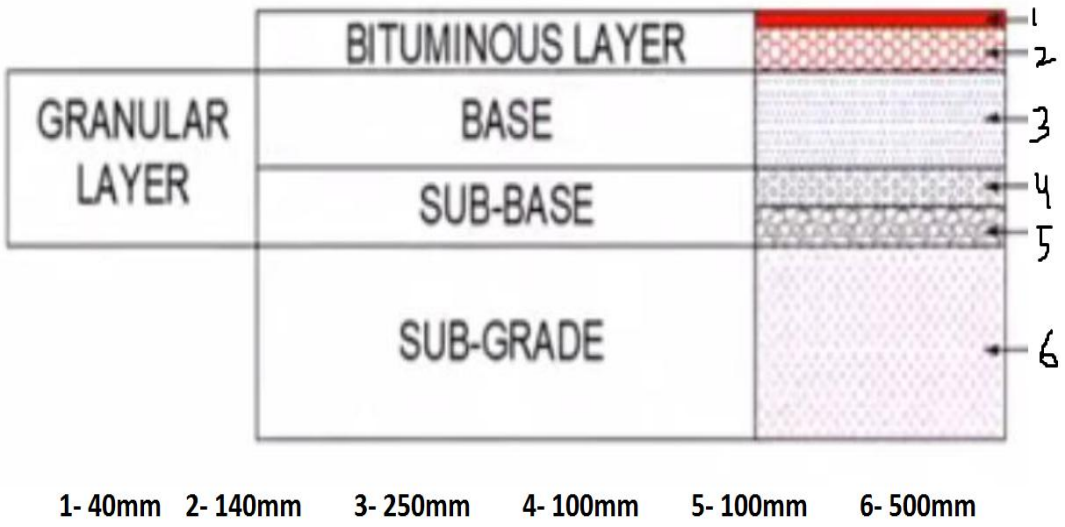


Figure 7.8: Adopted layer thickness

7. DESIGN AND ANALYSIS OF PAVEMENT

7.3.1.4 Failure Criteria and Strain Calculation

Reliability: 90% Reliability for all important roads such as expressways , national highways, state highways, urban road and for other categories of road for design traffic 20MSA or more. 80% reliability for design traffic less than 20MSA.

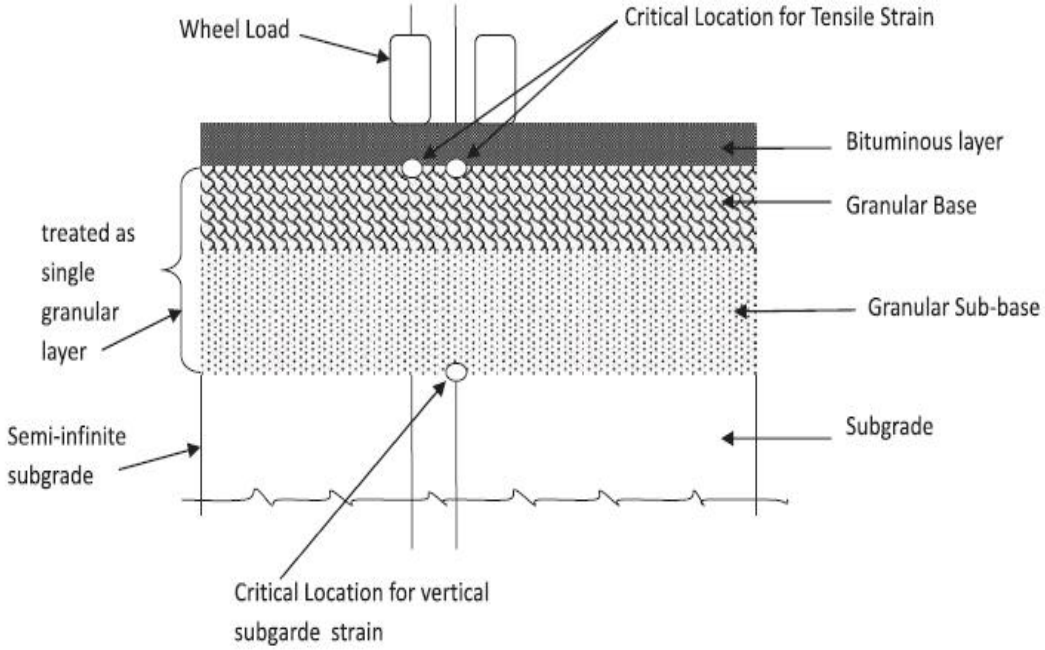


Figure 7.9: Critical Location of Strains

7. DESIGN AND ANALYSIS OF PAVEMENT

Rutting Criteria: The allowable number of load repetitions to control permanent deformation can be expressed as

$$N_R = 4.1656 \times 10^{-08} [1/\varepsilon_v]^{4.5337} \quad \text{(for 80 \% reliability)} \quad (3.1)$$

$$N_R = 1.4100 \times 10^{-08} [1/\varepsilon_v]^{4.5337} \quad \text{(for 90 \% reliability)} \quad (3.2)$$

Where

N_R = subgrade rutting life (cumulative equivalent number of 80 kN standard axle loads that can be served by the pavement before the critical rut depth of 20 mm or more occurs)

ε_v = vertical compressive strain at the top of the subgrade calculated using linear elastic layered theory by applying standard axle load at the surface of the selected pavement system

Fatigue Criteria:

Bituminous surfacings of pavements display flexural fatigue cracking if the tensile strain at the bottom of the bituminous layer is beyond certain limit. The relation between the fatigue life of the pavement and the tensile strain in the bottom of the bituminous layer was obtained as

$$N_f = 1.6064 * C * 10^{-04} [1/\varepsilon_t]^{3.89} * [1/M_{Rm}]^{0.854} \quad \text{(for 80 \% reliability)} \quad (3.3)$$

$$N_f = 0.5161 * C * 10^{-04} [1/\varepsilon_t]^{3.89} * [1/M_{Rm}]^{0.854} \quad \text{(for 90 \% reliability)} \quad (3.4)$$

Where

$C = 10^M$, and $M = 4.84 \left(\frac{Vbe}{Va+Vbe} - 0.69\right)$

V_a = per cent volume of air void in the mix used in the bottom bituminous layer

V_{be} = per cent volume of effective bitumen in the mix used in the bottom bituminous layer

N_f = fatigue life of bituminous layer (cumulative equivalent number of 80 kN standard axle loads that can be served by the pavement before the critical cracked area of 20 % or more of paved surface area occurs)

ε_t = maximum horizontal tensile strain at the bottom of the bottom bituminous layer (DBM) calculated using linear elastic layered theory by applying standard axle load at the surface of the selected pavement system

M_{Rm} = resilient modulus (MPa) of the bituminous mix used in the bottom bituminous layer, selected as per the recommendations made in these guidelines.

7. DESIGN AND ANALYSIS OF PAVEMENT

PERMISSIBLE STRAIN:

a) Vertical strain (90% reliability)

$N_R = 1.41 \text{ X } 10^{-8} [1/ \varepsilon_v]^{4.5337}$

$47 \text{ x } 10^6 = 1.41 \text{ x } 10^{-8} \text{ x } [1/ \varepsilon_v]^{4.5337}$

$\varepsilon_v = 0.0003768$

b) Horizontal strain (90% reliability):

$N_F = 0.561 \text{ X C X } 10^{-4} \text{ X } [1/ \varepsilon_t]^{3.89} \text{ x } [1/M_{RM}]^{0.854}$

$C = 10^M$ & $M = 4.84 [V_{be} / (V_{be} + V_a)]^{4.5337}$

Used, V_{be} =11.5 (Actual V_{be} : 11 to 13) & V_a = 3.5 (Actual V_a : not less than 3%)

M = 0.37 , C = 2.35

M_{RM} = 3000 Mpa (Figure 7.6)

$\varepsilon_t = 0.0001849095$

7. DESIGN AND ANALYSIS OF PAVEMENT

7.3.1.5 Modulus for each layer :

sub-grade modulus = 48.736 Mpa (From figure 7.4)

bitumen layer modulus = 3000 Mpa (From figure 7.6)

Sub-base usually consist of granular or WBM and the thickness should not be less than 150 mm for design traffic less than 10 msa and 200 mm for design traffic of 1.0 msa and above.

The modulus is found out by Granular layer modulus = $M_{GRAN} = 0.2\ (h)^{0.45} \times M_{support}$

h = Thickness of the DBM layer

$M_{GRAN} = 0.2\ (250+200)^{0.45} \times 48.736 = 152.344$ Mpa

Granular layer modulus = 152.344 Mpa

7. DESIGN AND ANALYSIS OF PAVEMENT

7.3.1.6 Actual strain by IITPAVE

Inputs:
Wheel load = 20000 N

Tyre Pressure = 0.56 MPa

Poissons's Ratio = 0.35

Radial distance = 155mm

Strain Comparision:

1. Actual horizontal strain < = Permissible horizontal strain

2. Actual vertical strain < = Permissible vertical strain

7. DESIGN AND ANALYSIS OF PAVEMENT

Trail – 1:

SURFACE COURSE = 30mm, BASE / BINDER COURSE = 100mm, WMM = 250mm, GSB = 200mm, SUBGRADE = 500mm

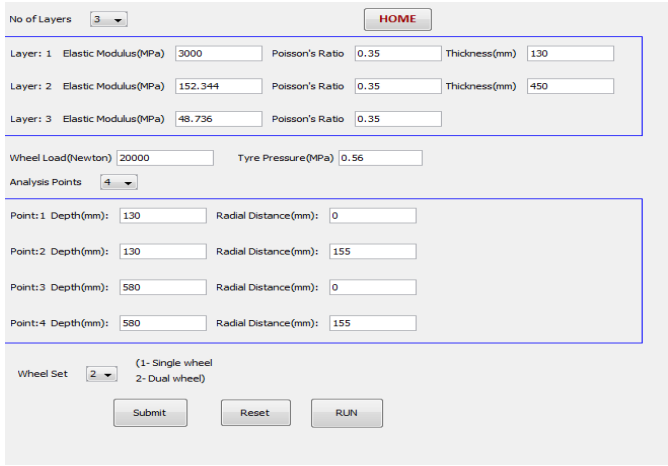


Figure 7.10: Trail 1 Input

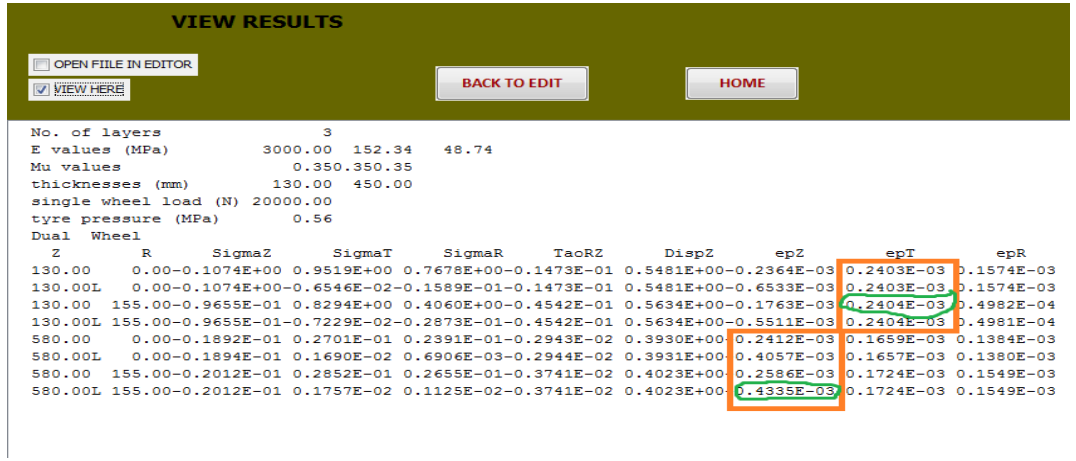


Figure 7.11: Trail 1 Output

Strain Comparision:

Permissible horizontal strain = 0.0001849095 Actual horizontal strain = 0.0002404

Permissible vertical strain = 0.0003768 Actual vertical strain = 0.0004335

Here, 1. Actual horizontal strain > Permissible horizontal strain

2. Actual vertical strain > Permissible vertical strain **(Hence Pavement is unsafe)**

7. DESIGN AND ANALYSIS OF PAVEMENT

Trail – 2:

SURFACE COURSE = 30mm, BASE / BINDER COURSE = 150mm, WMM = 250mm, GSB = 200mm, SUBGRADE = 500mm

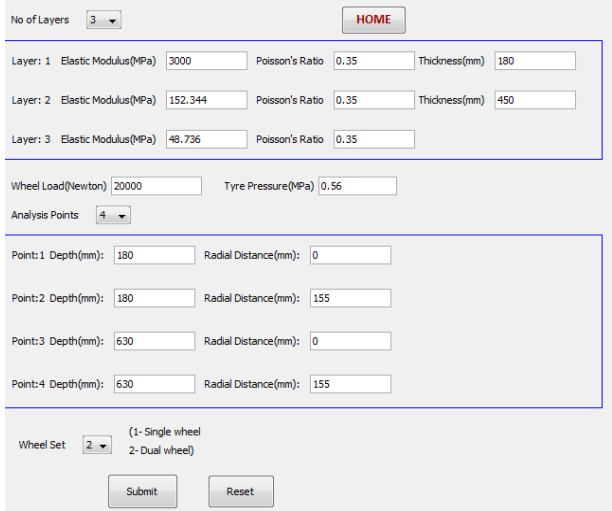


Figure 7.12: Trail 2 Input

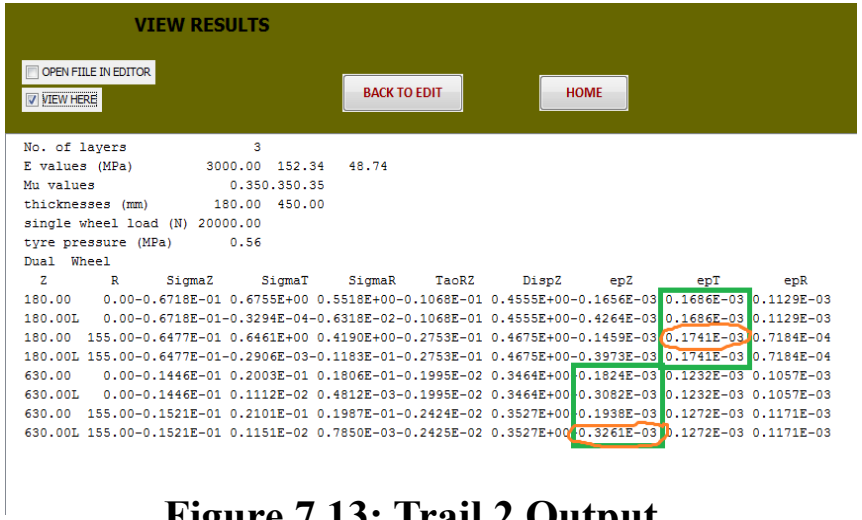


Figure 7.13: Trail 2 Output

Strain Comparision:

Permissible horizontal strain = 0.0001849095 Actual horizontal strain = 0.0001741

Permissible vertical strain = 0.0003768 Actual vertical strain = 0.0003261

Here, 1. Actual horizontal strain < Permissible horizontal strain

2. Actual vertical strain < Permissible vertical strain **(Hence Pavement is Safe)**

7. DESIGN AND ANALYSIS OF PAVEMENT

7.3.1.7 Selection of Economicdesign composition: With try of various thickness combination one can compare and compute the economical option for pavement structure.

Table.7.1 Economic pavement composition

Trail	Surface course (mm)	Binder Course (mm)	WMM(mm)	GSB (mm)	Subgrade (mm)	Permissible strain		Actual strain		Safe / unsafe
						Horizontal	Vertical	Horizontal	Vertical	
1	30	100	250	200	500	0.0001849095	0.0003768	0.0002404	0.0004355	unsafe
2	30	150	250	200	500			0.0001741	0.0003261	safe
3	30	150	200	200	500			0.0001762	0.0003590	safe
4	30	120	200	200	500			0.0002131	0.0004286	unsafe
5	30	130	200	200	500			0.0001998	0.0004035	unsafe
6	30	140	200	200	500			0.0001875	0.0003803	unsafe

8. CONCLUSIONS

- The thickness of pavement varies with the change in the value of C.B.R. With higher value of C.B.R. the pavement thickness is less and vice versa. In this study we used 90th percentile existing soil CBR value 4.6% & Borrow soil CBR value 8.5%.

- The average daily traffic is obtained from the traffic survey data: average traffic volume National Highway NH365BB (Suryapeta – Khammam) with chainage length of 10km from Mothey, Suryapet to Nayakulagudem, Kothagudem Telangana is 2460 CVPD.

- The Pavment layer thickness for construction of 6-lane flexible pavement as per IRC 37 -2018 designed those thickness are ; SURFACE COURSE = 30mm, BASE / BINDER COURSE = 150mm, WMM = 200mm, GSB = 200mm, SUBGRADE = 500mm. Due to the saving in Pavement thickness is less quantity of material will be applicable so that, huge amount of money can be saved

FUTURE SCOPE

- The present study is made on one road and the pattern of deterioration is determined on basis of the analysis. However, the more realistic pattern of deterioration can be developed by studying the different road.
- The study can also be with different bitumens grades, modified bitumen, RAP utilization.
- The study can also be elaborated by varying the various factors such as temperature and Rainfall Data.
- Further this research work can be carried with different materials to improve CBR values by soil stabilization technique.

REFERENCES

[1] Harish G R(4) (2017), “Analysis of Flexible Pavements Design Using IIT PAVE”, Imperial Journal of Interdisciplinary Research (IJIR), Vol-3, Issue-6, 2017, ISSN 2454 1362.
[2] Komershetty Gouthami, Dr KSR Murthy, D. Sandya Rani (7) (2017), “Flexible pavement design and Comparison of alternative Pavements using IRC 37-2012 IIT PAVE”, International journal for scientific Research and Development, vol-5, Issue-6, 2017, ISSN: 2321-0613.
[3] S S Jain, Sanjiv Aggarwal and M Parida (2005) “HDM-4 Pavement Deterioration Models for Indian National Highway Network”, Journal of Transportation Engineering, Volume 131 Issue 8 – August 2005.
[4] Dattatraya Thube and Aruna Thube (2013) “Software Development for Calibration of Highway Development and Management Tool (HDM-4) for Local Conditions”, International Journal of Structural and Civil Engineering Research, India.
[5] IRC: 37-2012, “Guidelines for the design of flexible pavements”, Indian Road Congress, New Delhi, 2012.
[6] Highway Development and Management Model HDM-4 Manual.
[7] Highway Engineering- S. K. Khanna & C. E. G Justo and A.Veeraragavan, Revised 10th Edition.
[8] Whiteman, D.C., Eric, E.S., and Dakin, J M., “Software User Guide”, Vol. 3, HDM-4 Documentation Series, Permanent International Association of Road Congresses (PIARC) -World Road Association, France, 2000.
[12] CRRI (1994) “Pavement Performance Study on Existing Pavement Sections”, Project sponsored by Ministry of Surface Transport, Government of India, Final
Report, Submitted by Central Road Research Institute, New Delhi.

THANK YOU